

Interpreting the Solution of a System on a Graph

Answer Key

Algebra 1

Quick Answers

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| 1. $x = 2, y = 3$ | 6. (a) second equation solved for $y = 4 - 2x$, — |
| 2. $x = 3, y = 5$ | 7. $x = 3, y = 2$ |
| 3. (a) number of solutions = $\emptyset$, — | 8. (a) solution = $x = 4, y = 4$, — |
| 4. $x = 4, y = 2$ | 9. (a) solution = $x = 0, y = 3$, — |
| 5. $x = 4, y = 60$ | 10. (b) break-even point = $x = 5, y = 50$, — |
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What is verified

5 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 6, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-REI.C.6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 15 tagged checks.

1. $A: y = x + 1, B: y = -x + 5$. Solution from the graph.

Solution: The two lines cross at the lattice point where $x = 2$: line A gives $y = 2 + 1 = 3$ and line B gives $y = -2 + 5 = 3$, so both agree there. That point, and only that point, sits on both lines. (2, 3)

2. $A: y = 2x - 1, B: y = x + 2$. Solution from the graph.

Solution: Reading the crossing point off the grid gives $x = 3, y = 5$. Algebra confirms it: $2x - 1 = x + 2$ gives $x = 3$, and then $y = 3 + 2 = 5$. (3, 5)

3. $A: y = 2x + 1, B: y = 2x - 3$. (a) How many solutions? (b) Why?

Solution (a): The lines never meet, so no ordered pair lies on both. no solution

Solution (b) — instructor-judged. Both equations have slope 2, so the lines climb at exactly the same rate and stay a constant vertical distance of 4 apart forever; their y -intercepts differ (1 and -3), so they are not the same line either. Setting the right-hand

sides equal gives $1 = -3$, which is false — the algebra says the same thing the picture does. Full credit links equal slopes and different intercepts to the absence of a crossing point.

4. Graph $y = -x + 6$ and $2y = x$; give the solution.

Solution: The first line has y -intercept 6 and slope -1 . The second is $y = \frac{1}{2}x$, through the origin with slope $\frac{1}{2}$. They cross where $-x + 6 = \frac{1}{2}x$, that is $6 = \frac{3}{2}x$, so $x = 4$ and $y = 2$. On the grid the crossing sits at a lattice point, which is how a graphical reading can be trusted here.

(4, 2)

5. Gym plans $A: y = 10x + 20$, $B: y = 5x + 40$. Equal-cost point.

Solution: The lines cross where $10x + 20 = 5x + 40$, so $5x = 20$ and $x = 4$ months; substituting back, $y = 10(4) + 20 = 60$ dollars. On the graph this is the one place where the steeper plan A line meets plan B .

(4, 60)

Reading it in context: before 4 months plan A is cheaper, after 4 months plan B is.

6. $y = -2x + 4$ and $4x + 2y = 8$: (a) solve the second for y ; (b) how many solutions?

Solution (a): Subtract $4x$ from both sides: $2y = 8 - 4x$. Divide both sides by 2: $y = 4 - 2x$.

$y = 4 - 2x$

Solution (b) — instructor-judged. That is the first equation written the other way round, so the two equations describe one and the same line — which is why only one line appears on the graph. Every point on that line satisfies both equations, so the system has infinitely many solutions. Full credit says the lines coincide and that every point on the line is a solution.

7. Graph $2x + y = 8$ and $y = x - 1$; give the solution.

Solution: Rewrite the first as $y = -2x + 8$ (intercept 8, slope -2); the second has intercept -1 and slope 1. Setting them equal: $-2x + 8 = x - 1$, so $9 = 3x$ and $x = 3$, giving $y = 3 - 1 = 2$. Check in the untouched first equation: $2(3) + 2 = 8$.

(3, 2)

8. $A: 2y = x + 4$, $B: y = -x + 8$. (a) the solution; (b) why $(6, 5)$ is not one.

Solution (a): Line A is $y = \frac{1}{2}x + 2$. Setting the two right-hand sides equal, $\frac{1}{2}x + 2 = -x + 8$, so $\frac{3}{2}x = 6$ and $x = 4$; then $y = -4 + 8 = 4$.

(4, 4)

Solution (b) — instructor-judged. $(6, 5)$ does satisfy line A : $2(5) = 6 + 4$. But a solution of a *system* has to satisfy both equations at the same time, and in line B it gives $5 \neq -6 + 8 = 2$. On the graph the point sits on one line and well away from the other. Full credit uses the both-equations-at-once idea.

9. $A: y = 2x + 3$, $B: y = -x + 3$. (a) the solution; (b) predicting it.

Solution (a): The lines cross on the y -axis.

(0, 3)

Solution (b) — instructor-judged. Both equations have the same constant term, 3, so both lines pass through $(0, 3)$ — they already share that point before any graphing is done. Their slopes, 2 and -1 , are different, so they cannot share any other point; two distinct lines meet at most once. Full credit refers to the shared intercept and the different slopes.

10. Van plans A : $y = 4x + 30$, B : $y = 8x + 10$. (a) graph; (b) break-even; (c) which is cheaper after it?

Solution (a): Plan A starts at 30 dollars and rises 4 dollars per day; plan B starts at 10 dollars and rises 8 dollars per day. Plotting the intercepts and using the slopes gives two lines that cross inside the grid.

Solution (b): $4x + 30 = 8x + 10$ gives $20 = 4x$, so $x = 5$ days, and $y = 4(5) + 30 = 50$ dollars. Both plans cost 50 dollars for a five-day rental. $(5, 50)$

Solution (c) — instructor-judged. To the right of the crossing point the plan B line, being steeper, sits above the plan A line, and higher on this graph means more dollars. So for rentals longer than 5 days plan A costs less; for shorter rentals plan B does. Full credit reads the comparison off the graph as which line is on top after the crossing.